Bubble Transformer V5: Focus-Inspired Sinkhorn Token Grouping for Causal Language Model Attention

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Abstract

We present Bubble Transformer V5 (BT V5), a hybrid attention mechanism that uses Sinkhorn normalization for token grouping while preserving standard softmax within groups. Unlike previous BT V5 variants that replaced softmax with doubly-stochastic attention—destroying the peaked distribution needed for language modeling—BT V5 restricts Sinkhorn's role to grouping tokens, then applies standard softmax to maintain peakedness. Through single-layer frozen swap experiments on Qwen3-0.6B with WikiText-2, we find that: (1) Focus Bubble passes the $\leq 2\%$ PPL gate at multiple layers, (2) the optimal configuration is Layer 7 with $\epsilon = 0.001$, $\tau = 1$, $\lambda = 0.3$ achieving PPL$= 22.550$ ($+0.16\%$), and (3) FocusDeltaNet (combining Focus grouping with DeltaNet delta rule) provides $4.7\times$ improvement over Focus-only. We present 501 tests passing and reproducible benchmark results. Multi-layer experiments show FocusDeltaNet scales to 3 layers at $+1.38\%$ PPL while pure Focus degrades with > 2 layers. Needle-in-a-Haystack evaluation at 2K tokens shows 100% retrieval accuracy for all configurations.

1 Introduction

1.1 Motivation

Previous BT V5 variants (V1–V4) attempted to replace softmax attention with doubly-stochastic attention (SIRI) or geometric cost functions. All variants failed the $\leq 2\%$ PPL gate on Qwen3-0.6B / WikiText-2, with best results ranging from $+2.39\%$ to $+211\%$ PPL drift.

The core problem: doubly-stochastic attention destroys the peaked distribution that softmax provides. Language modeling requires sharp attention distributions to concentrate on relevant tokens. Forcing doubly-stochastic normalization (row sums $= 1$, column sums $= 1$) distributes attention mass uniformly across all tokens, destroying peakedness.

1.2 The Insight from Focus

The Focus paper [1] demonstrated that Sinkhorn normalization can improve PPL when used for token grouping rather than attention normalization. The key idea:

1. Use Sinkhorn to create soft group assignments
2. Apply standard softmax *within* each group
3. This preserves peakedness while adding geometric structure

Focus achieved $42.8 \rightarrow 30.3$ PPL (29% improvement) on GPT-2 124M with this approach.

1.3 Contributions

This paper presents:

- The Focus Bubble architecture (Sinkhorn grouping + softmax within groups)
- Honest benchmark results with explicit failure modes
- Multi-layer scaling analysis (FocusDeltaNet scales to 3 layers)
- Engineering decisions and bug fixes for numerical safety

2 The PPL Gate

2.1 Gate Definition

Per BT V5 protocol:

$$\Delta\text{PPL} = \text{PPL}_{\text{BT}} - \text{PPL}_{\text{softmax}} \leq 2.00\% \tag{1}$$

For Qwen3-0.6B / WikiText-2 (50k chars, 44 windows):

- Baseline PPL: 22.513
- Gate max: 22.964

2.2 Previous Failures

Table 1: Previous BT V5 variants that failed the PPL gate.

Variant	Best PPL	Δ%	Gate
Pure SIRI (doubly-stochastic)	69.968	+211%	FAIL
Row-stochastic-only	28.601	+30%	FAIL
Hybrid Approach 1 (dot-product)	22.627	+2.85%	FAIL
Hybrid Approach 2 (hybrid cost)	22.627	+2.85%	FAIL
Hybrid Approach 3 (DeltaNet + SIRI bias)	23.052	+2.39%	FAIL
Learnable beta (Approach 3)	23.052	+2.39%	FAIL

3 Focus Bubble Architecture

3.1 Core Pipeline

FocusBubbleAttention (input: $[\mathrm{B}, \mathrm{N}, \mathrm{D}]$):

1. **Q, K, V projections**: Standard learned projections
2. **Compute scores**: $S = QK^T/\sqrt{d}$ (dot-product, NOT geometric)
3. **Add Power Diagram bias**: $S = S + \psi$ (learnable per-head)
4. **Sinkhorn for grouping**: Apply τ iterations of Sinkhorn-Knopp normalization
5. **Softmax within groups**: Standard softmax on grouped scores (preserves peakedness)
6. **Output projection**: Standard learned output projection

3.2 Key Difference from SIRI

Table 2: Comparison of SIRI (V1–V4) vs Focus Bubble (V5).

Component	SIRI (V1–V4)	Focus Bubble (V5)
Sinkhorn role	Normalize attention weights	Group tokens
Softmax	Replaced by doubly-stochastic	Preserved within groups
Peakedness	Destroyed	Preserved
PPL impact	+30% to +211%	+0.16% to +0.74%

3.3 Mathematical Formulation

Let $S \in \mathbb{R}^{N\times N}$ be the score matrix, $\psi \in \mathbb{R}^{H}$ be the per-head bias.

Sinkhorn grouping (τ iterations):

$$S_0 = S \tag{2}$$

$$S_k = \mathrm{log\_sinkhorn}(S_{k-1}) \tag{3}$$

$$\mathrm{groups} = \exp(S_\tau) \tag{4}$$

Softmax within groups:

$$\mathrm{attn} = \mathrm{softmax}(S + \log(\mathrm{groups}), \mathrm{dim} = -1) \tag{5}$$

$$\mathrm{output} = \mathrm{attn} \cdot V \tag{6}$$

3.4 FocusDeltaNet Variant

For the best results, we combine Focus grouping with DeltaNet delta rule:

$$\mathrm{output} = \lambda \cdot \mathrm{DeltaNet}(Q, K, V) + (1 - \lambda) \cdot \mathrm{FocusBubble}(Q, K, V) \tag{7}$$

With safe normalization of Q, K, V (norm clamping to prevent overflow), this achieves PPL= 22.550 at $\lambda = 0.3$.

4 Experimental Results

4.1 Setup

- Model: Qwen3-0.6B-Base (float16, eager attention)
- Dataset: WikiText-2 test split, 50k chars
- Hardware: GTX 1650 (4.3GB VRAM)

4.2 Parameter Sensitivity (L7, Focus Only)

Epsilon sweep ($\tau = 5$):

Tau sweep ($\epsilon = 0.001$):

Table 3: Epsilon sweep at $\tau = 5$ (all fail).

Epsilon	PPL	Gate
0.001	23.034	FAIL
0.010	23.039	FAIL
0.100	23.082	FAIL
1.000	23.606	FAIL

Table 4: Tau sweep at $\epsilon = 0.001$.

τ	PPL	Gate
1	**22.706**	**PASS**
2	22.843	PASS
3	22.930	PASS
5	23.034	FAIL

4.3 Layer Sweep ($\epsilon = 0.001$, $\tau = 1$)

4.4 FocusDeltaNet Lambda Sweep

4.5 Multi-Layer Scaling

4.6 Needle-in-a-Haystack (NIAH)

5 L9 Anomaly

L9 is the only mid-layer that fails the gate (+2.28%) while neighbors L10/L11/L12 pass comfortably (+1.08%, +1.15%, +0.86%). Re-measurement confirms exact replication (diff = 0.0002 PPL).

Per-head analysis shows L9's outlier heads (H2: 131.81%, H6: 153.63%) are not geometric outliers compared to L10's outliers (H6, H9). The failure appears to stem from integration effects across heads, not individual head geometry.

6 Power Diagram ψ: Honest Caveat

The ψ bias is **absorbed by Sinkhorn column normalization**. Empirically:

- L12 FocusOnly psi=True: PPL= 23.082
- L12 FocusOnly psi=False: PPL= 23.082
- **Identical results** — ψ has no effect on output

7 Numerical Safety Engineering

Key bugs fixed:

1. **FocusDeltaNet NaN**: Missing Q/K/V normalization in delta rule — fixed with `_safe_normalize()`
2. **q_norm shape**: Applied view+transpose BEFORE q_norm for head-wise RMSNorm

Table 5: Layer sweep results.

Layer	PPL	Gate
L3	22.960	PASS
L5	22.928	PASS
L7	**22.681**	**PASS**
L9	23.026	FAIL
L10	22.757	PASS
L11	22.772	PASS
L12	22.706	PASS
L15	22.814	PASS
L19	22.850	PASS
L23	23.154	FAIL

Table 6: Lambda sweep at L7, $\epsilon = 0.001$, $\tau = 1$.

λ	PPL	Gate
0.0 (Focus only)	22.681	PASS
0.2	22.554	PASS
0.3	**22.550**	**PASS**
0.5	22.651	PASS
0.7	22.900	PASS
0.8	23.079	FAIL
0.9	23.297	FAIL
1.0 (DeltaNet only)	23.558	FAIL

3. **RoPE signature**: Corrected to `apply_rotary_pos_emb(Q, K, cos, sin)`
4. **Wrapper return type**: Changed to 2-tuple (out, attn_weights)

8 Reproducibility

- Code: `https://github.com/Markush418/LLM-BUBBLE-TRANSFORMER`
- Tests: 501 passing, 0 failed
- Hardware: GTX 1650 (4.3GB VRAM)
- Time: ∼30 minutes for full benchmark suite

9 Limitations and Future Work

Limitations:

- Single-layer swap only (not validated with pretrain from scratch)
- Limited long-context (NIAH at 2K tokens, GTX 1650 VRAM limit)
- No downstream evaluation (MMLU, HumanEval, GSM8K)

Table 7: Multi-layer swap results.

Configuration	PPL	Δ%	Gate
Focus L7	22.681	+0.74%	PASS
Focus L7+L10	23.100	+2.61%	FAIL
Focus L7+L12	22.883	+1.64%	PASS
Focus L7+L10+L12	23.339	+3.67%	FAIL
FocusDeltaNet L7	22.550	+0.16%	PASS
FocusDeltaNet L7+L10	22.648	+0.60%	PASS
FocusDeltaNet L7+L10+L12	22.825	+1.38%	PASS

Table 8: NIAH retrieval accuracy at 2K tokens.

Configuration	Retrieval Accuracy
Softmax Baseline	100%
Focus Bubble L7	100%
FocusDeltaNet L7	100%

- L9 anomaly unexplained

Future Work:

- Pretrain Bubble-1.3B from scratch
- Long-context evaluation (4K–32K tokens)
- Downstream task evaluation
- Speed optimization (CUDA kernel)

References

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[3] Kimi Linear: Hybrid linear attention with channel-wise decay. arXiv:2510.26692, 2025.

[4] Qwen3 Technical Report. arXiv:2505.09388, 2025.